Ashwin Parameswaran

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OBJECTIVE

Driven computer science student passionate about AI/ML, data, and software engineering in physical systems. Experienced in applying intelligent models to domains like open-source intelligence, autonomous vehicles, control systems, and embedded platforms. Skilled in research-driven problem solving, technical communication, and development for high-impact engineering.

EDUCATION

Texas A&M University, College Station, TX – *College of Engineering, Class of '26*
Junior Honors Student, Pursuing BS in Computer Science w/ Minors in Electrical Engineering & Mathematics
GPA: 3.701
Financed 100% of college education through merit scholarship awards.
Uplift North Hills Preparatory, Irving, TX – *Class of '22*
International Baccalaureate Diploma
Weighted GPA: 4.61 **-** Ranked 3rd of Graduating Class

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, PostgreSQL, React.js, Assembly, Ladder Logic, HTML/CSS, Javascript, React.js
- **AI/ML:** Transformers, LLMs, CV/NLP, CUDA, PyTorch, AWS, Zero-Trust Inference, Secure Model Deployment
- **Embedded Systems:** Arduino, STM32/Nucleo, CAN, RT Logic, PCB Design (Altium), ROS, Linux
- **Tools:** Git, Docker, OpenCV, Linux/Unix, ICS/SCADA, Bash, SLAM, VSCode

EXPERIENCE

Griffiss Institute – *VICEROY MAVEN Intern:* June 2024 - Present
- Co-authored & co-developed VEIL, (Veiled Extraction of Information from LLMs) for geolocation for DoD open-source intelligence gathering and operations research.
- Designed and authored *MultiSurf*, an DL-powered video analysis pipeline that performs multimodal embedding (CLIP, YAMNet, Sentence-BERT) and semantic video segmentation for public info & intelligence gathering.

Texas A&M Sketch Recognition Lab – *Research Project:* June 2024 - Present
- Improved convolutional vision models for medical imaging via segmentation/classification tuning and architecture.
- Brought training time down by 3x and improved model accuracy by an average of 2% across five datasets.
- Collaborated on reproducibility testing and optimization of cutting-edge CV research.

Institute for a Disaster Resilient Texas – *Research Project:* September 2024 – Present
- Built GenAI pipelines for classifying disaster imagery using multimodal models (NLP + CV).
- Developed scalable image-to-text analytics for emergency response and impact mapping.
- Co-authored in research poster for computer vision-based elevation estimation for disaster prediction.

Texas A&M Cybersecurity Lab – *ICS Engineer* Nov 2023 - Present
- Engineered PLC control logic and anomaly detection models for industrial systems in simulated OT/IT environments
- Developed scenes and control logic for OT environments, researched ML for anomaly detection

ENDEAVR Institute – *Autonomous Driving Research Intern:* October 2024 – March 2025
- Developed Python scripts for real-time video capture and preparation, enabling remote autonomous driving
- Part of an agile team converting legacy vehicles into remotely monitored autonomous platforms
- Contributing to a scalable, remotely monitored autonomous rescue and emergency service vehicle platform.

ACTIVITIES

Study Abroad: Swansea University, UK Jan 2025 – Jun 2025
- Completed CS/Math Coursework while immersed in UK academic paradigm
- Gained international perspective and adaptability through independent travel and academic success abroad

Texas A&M Solar Car Racing Team Oct 2022 - Present
- Worked with CAN bus, Microelectronic Software, PCB Design & Verification for endurance racing.
- Developed & implemented control systems & algorithms for vehicle strategy using car & environmental data
- Finished 7th of 42 teams in the American Solar Challenge & Formula Sun Grand Prix 2024

VICEROY Scholar Jan 2024 - Present
- Virtual Institutes for Cyber and Electromagnetic Spectrum Research and Employ
- Received stipends for research & coursework in data science, cryptography, and cybersecurity